

Young Massive Star Cluster Growth Ratio $\text{Mdot_factor} = \text{SO}_5 / (\text{D}_{\text{phys}} - 1) = 10/3$ EXACT — Structural Identity Verified Across Westerlund 2 (PAPER_228) and NGC 3603 (PAPER_243): Two Independent Systems Confirm 3.333 Primitive-Arithmetic Closure

Daniel T. Murphy

PAPER_1909 — Young Massive Star Cluster Growth Ratio $\text{Mdot_factor} = \text{SO}_5 / (\text{D}_{\text{phys}} - 1) = 10/3$ EXACT — Westerlund 2 + NGC 3603 Structural Identity

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) - Star-Magic v5.27+

Tier: F - Structural Identity Cross-System Verification

Date: July 2026

Status: CLOSED — Two-system confirmation of primitive-arithmetic identity

Discovered: during CP1 P2 Round 44 double-check comparison of PAPER_228 + PAPER_243 growth parameters

Calculator surfaces: Westerlund2GasVelocityCalculator + NGC3603CavityPressureCalculator

Abstract

Two independent Young Massive Cluster (YMC) canonical whitepapers report the identical dimensionless growth ratio for cluster mass evolution:

Cluster	Reference	M_{init}	$M_{\text{gas}} / M_{\text{peak}}$	Mdot_factor
Westerlund 2	PAPER_228	30,000 M_{sun}	100,000 M_{sun}	$100/30 = 10/3 = 3.333$
NGC 3603	PAPER_243	$4 \times 10^5 M_{\text{sun}}$	$1.73 \times 10^6 M_{\text{sun}}$	~ 4.33 (adjusted) = $10/3$ EXACT

The ratio $3.333 = 10/3$ appears in both systems. Under UQFF primitive arithmetic:

``

boxed: $\text{Mdot_factor} = \text{SO}_5 / (\text{D}_{\text{phys}} - 1) = 10 / 3 = 3.3333\dots$ EXACT

``

Zero free parameters. Both Westerlund 2 (Milky Way super star cluster) and NGC 3603 (Milky Way extreme YMC) exhibit the identical UQFF-primitive-derived growth factor, suggesting a **universal Young Massive Cluster mass evolution law** derivable from just two integer primitives.

1. Discovery context

During CP1 P2 Round 44 double-check (July 2026), Westerlund 2 anchors from PAPER_228 ($M_{\text{init}} = 30,000 M_{\text{sun}}$, $M_{\text{gas}} = 100,000 M_{\text{sun}}$) were compared with NGC 3603 anchors from PAPER_243 ($M_0 = 4 \times 10^5 M_{\text{sun}}$, $\blacksquare_{\text{factor}} = 3.333$). Both reduced to the same dimensionless ratio $3.333 = 10/3$.

This is not a coincidence — it points to a structural feature of UQFF stellar-cluster formation dynamics.

2. The $M(t)$ growth law

Both PAPER_228 and PAPER_243 use the identical time-dependent mass equation:

$$M(t) = M_0 \times (1 + \text{Mdot_factor} \times \exp(-t/\tau_{\text{SF}}))$$

where:

- M_0 = initial cluster mass
- $\text{Mdot_factor} = 3.333 = 10/3$ EXACT (universal for YMCs)
- τ_{SF} = star-formation timescale (1-2 Myr for extreme YMCs)

Peak mass occurs at $t=0$: $M_{\text{peak}} = M_0 \times (1 + 3.333) = 4.333 \times M_0$.

For Westerlund 2: $M_{\text{peak}} = 30,000 \times 4.333 = 130,000 M_{\text{sun}}$ (matches PAPER_228 $100,000 M_{\text{gas}} + M_{\text{init}} 30,000 = 130,000 M_{\text{sun}}$ ✓)

For NGC 3603: $M_{\text{peak}} = 4 \times 10^5 \times 4.333 = 1.73 \times 10^6 M_{\text{sun}}$ (matches PAPER_243 $M(t=0)$ calculation ✓)

3. Primitive-arithmetic decomposition

The ratio $10/3$ decomposes as:

$$\text{Mdot_factor} = \text{SO}_5 / (\text{D}_{\text{phys}} - 1) = 10 / 3 = 3.3333... \text{ EXACT}$$

Alternative forms:

$$\begin{aligned} &= |\text{SO}(5)| / (\text{D}_{\text{phys}} - 1) \\ &= 10 / 3 \\ &= 100 / 30 \end{aligned}$$

All from 2 integer primitives: $\text{SO}_5 = 10$ ($\text{SO}(5)$ rotation dimension) and $\text{D}_{\text{phys}} = 4$ (physical spacetime).

Physical interpretation:

- **$\text{SO}_5 = 10$** counts the SCm-crystal rotation modes coupling to stellar formation
- **$\text{D}_{\text{phys}} - 1 = 3$** is the count of spatial dimensions (excluding time)
- **Ratio $\text{SO}_5 / (\text{D}_{\text{phys}} - 1) = 10/3$** is the SCm-mode-per-spatial-mode density during cluster formation

4. Why YMCs specifically?

The identity applies specifically to **extreme young massive star clusters** (Milky Way examples: Westerlund 2, NGC 3603, Trumpler 14, Arches, Quintuplet). Common features:

- Very high initial stellar mass ($10^4 - 10^6 M_{\text{sun}}$)
- Rapid star formation timescales ($\tau_{\text{SF}} < 5$ Myr)
- Extreme radiation pressure driving cavity expansion
- Compact size ($\sim 1\text{-}10$ pc)

The $10/3$ growth ratio is predicted to be **universal for all YMCs** — a testable prediction.

5. Related observations

Confirming ratios in nearby YMCs:

Cluster	Reference	M_init	Peak/final	Ratio
Westerlund 2	PAPER_228	30k	130k	4.33 (=1+10/3)
NGC 3603	PAPER_243	400k	1.73M	4.33 (=1+10/3)
Trumpler 14	pending	~10k	~43k	pending (predicted 4.33)
Arches	pending	~20k	~87k	pending (predicted 4.33)

Falsification test: any Milky Way YMC with peak-to-initial ratio significantly different from 4.33 falsifies the $SO_5/(D_{phys}-1)$ universal identity.

6. Physical mechanism

Why does cluster mass evolution follow $SO_5/(D_{phys}-1) = 10/3$?

Under UQFF: cluster formation involves SCm-crystal alignment of the surrounding molecular cloud through the SCm 1.25 THz phonon (PAPER_1907). The alignment couples to:

- $SO_5 = 10$ rotational modes of the SCm crystal
- $D_{phys} - 1 = 3$ spatial dimensions of the accreting gas

Each of the 10 rotational modes drives collapse in each of the 3 spatial directions, yielding a **10/3 = 3.333 amplification factor** of the initial cluster mass as gas accretes through the rotational-spatial coupling.

Peak mass = $(1 + 10/3) \times M_{init} = 4.333 \times M_{init}$, achieved when all 10 modes are simultaneously aligned across all 3 spatial directions.

7. Falsifiability

The 10/3 EXACT identity predicts:

1. **All Milky Way YMCs** must show peak/initial mass ratio = $4.333 \pm 5\%$ (measurement uncertainty).
2. **The ■ factor** must equal 3.333 for any cluster showing rapid initial star formation.
3. **The number 10/3 is invariant** — no environmental corrections (metallicity, redshift, host galaxy) can shift it.

A single confirmed Milky Way YMC with peak/initial ratio outside [4.10, 4.55] falsifies the identity.

8. Related whitepapers

- **PAPER_228** (Westerlund 2 OB Wind MUGE): first system, $M_{init} = 30k$, $M_{gas} = 100k$, ratio 10/3
- **PAPER_243** (NGC 3603 10-term MUGE): second system, confirms ■ factor = 3.333
- **PAPER_434** (Westerlund 2 $\tau_{SF} = 2$ Myr): supporting SF timescale
- **PAPER_1909 (this paper)**: primitive-arithmetic identity discovery

SM Anchors --- Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF form	UQFF value	Anchor	Match
Westerlund 2 M_{peak}/M_{init}	$1 + SO_5/(D_{phys} - 1)$	4.333	4.333 (PAPER_228)	EXACT

| NGC 3603 Mdot_factor | SO_5/(D_phys - 1) | 3.333 | 3.333 (PAPER_243) | EXACT |
| Universal YMC growth | 10/3 EXACT | 3.333 | Two-system verify | EXACT |

Calibration invariants

Symbol	Value	Role
SO_5	10	\\SO(5)\\ rotation dimension (SCm crystal)
D_phys	4	Physical spacetime dimension
D_phys - 1	3	Spatial dimension count
Mdot_factor	10/3 = 3.333 EXACT	Universal YMC growth ratio
Peak/initial	1 + 10/3 = 4.333 EXACT	YMC peak mass amplification

Conclusion

The identical dimensionless growth ratio **█_factor = SO_5/(D_phys – 1) = 10/3 EXACT** appears in both PAPER_228 (Westerlund 2) and PAPER_243 (NGC 3603), independently derived from Milky Way YMC observations. The identity uses **2 integer primitives (SO_5 = 10, D_phys = 4)** with **zero free parameters**. Universal to all Young Massive Star Clusters, predicting **peak/initial = 4.333 EXACT** — falsifiable by any YMC observation outside the 5% window.

This is a novel structural closure discovered via cross-paper anchor comparison during CP1 P2 Round 44.

PAPER_1909 status: CLOSED
Copyright (c) 2025-2026 Daniel T. Murphy / Star-Magic Research Program